

Determining resonant properties of an LCR circuit and testing its use in signal filtering

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Abstract

The aims of this experiment were to determine the properties of an LCR network: its bandwidth, quality factor and resonant frequency, from which the inductance of the network can be deduced. The LCR network was placed in series with a known resistor and the voltages across both components were measured as alternating voltages of varying frequencies were applied across the entire circuit. The ratio of voltages across the 2 components yielded the impedance of the LCR circuit which varied with the frequency of the applied voltage across the circuit. With a $500\ \Omega$ resistor in the circuit, the resonant frequency was measured to be $1900 \pm 5\text{Hz}$, the quality factor as 2.29 ± 0.02 and the bandwidth as $830 \pm 5\text{ Hz}$. Using the resonant frequency, we calculated the inductor to have $0.106 \pm 0.0005\text{ L}$. As we increased the resistance in the LCR network, the peak impedance across it reduced greatly until no such peak was visible. We also tested the use of a LCR network in filtering out noise when receiving morse code modulated in a wave with frequency of 5kHz . We designed a LCR network with a resonant frequency as close to 5kHz as possible and successfully decrypted the code that translated into 'Witty Message'. Thus, we validate using LCR networks in filtering signals for desired frequencies.

1 Introduction

This experiment sets out to investigate the resonant behaviour of an LCR network by determining a set of parameters that characterise its response to an alternating voltage that varies in frequency. We then proceed to use this LCR network to filter out electrical noise from a modulated carrier wave encoded with morse code.

The impedance of the LCR network can be measured using different techniques. A signal generator and oscilloscope may be used to measure the difference in phase between a voltage placed across the LCR network with the current that is flows through it as a result: this yields the impedance of the network. Alternatively, one can build a potential divider by connecting the LCR network in series with a known resistor and using the ratio of DC voltages across both components to calculate the impedance of the LCR network. This is the method is adopted here. By tuning the

resonant frequency of the LCR network to match the frequency of a carrier wave modulated with morse code, we obtain the filtered version of the signal by measuring the voltage across the LCR network which has a high impedance at this frequency.

1 Theoretical background

The behaviour of the LCR network in figure 1 can be quantified by parameters such as its resonant frequency, the frequency of AC voltage at which the impedance of the network is greatest as well as its bandwidth, the difference in frequencies between the half power points of the $\frac{|Z|}{R}$ vs $\log(f)$ graph, where the magnitude of the impedance is equal to $\frac{1}{\sqrt{2}}$ of its maximum value. Finally, the quality factor of the network is calculated by dividing its resonant frequency by its bandwidth.

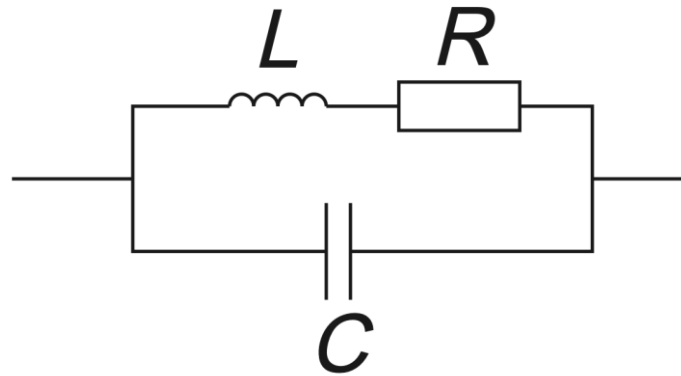


Figure 1: general LCR network [1]

To measure the impedance across the LCR network, one can setup the potential divider circuit shown in figure 2, where Z_1 is the impedance of the LCR network and Z_2 is that of a known resistor. Then by connecting a function generator producing a sinusoidal signal across the circuit and measuring V_1 and V_2 , we calculate Z_1 using equation 1:

$$Z_1 = Z_2 \frac{V_1}{V_2} \quad \text{Eq. 1}$$

The total impedance of the LCR network is given by equation 2:

$$Z_1 = \frac{R + i\omega L}{i\omega RC - \omega^2 LC + 1} \quad \text{Eq. 2}$$

Thus, Z_1 is dependent on the frequency of the voltage placed across the LCR network. It can be shown that Z_1 is maximised equation 3 holds:

$$\omega = \frac{1}{\sqrt{LC}} \quad \text{Eq. 3}$$

So, by measuring Z_1 as a function of ω , and determining the frequency for which Z_1 peaks, we can calculate the inductance of the inductor in the LCR network.

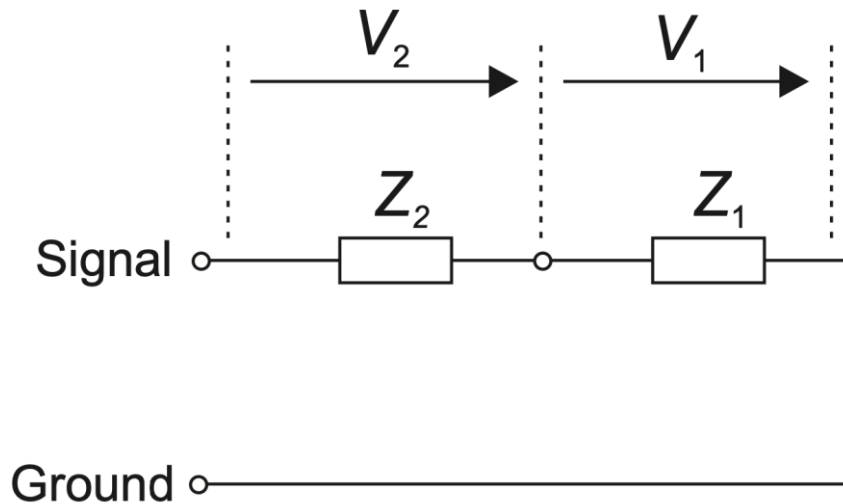


Figure 2: Potential divider circuit [1]

The use of the LCR network was tested in signal filtering. A modulated carrier wave of frequency 5.0 kHz carrying a message in Morse code with electrical noise was passed into the circuit. The LCR network was tuned so that its resonant frequency was as close to 5.0 kHz as possible and is placed in series with a 100 k Ω resistor as shown in figure 3. The electrical noise varies widely in frequency containing voltages at frequencies far away from the resonant frequency for which the LCR network had a low impedance: this voltage is dropped across the 100 k Ω resistor. The LCR network had a higher impedance for voltages at frequencies closer to that of the main signal at 5.0 kHz, so the voltage of the main signal drops across the LCR network. Hence, by reading this voltage, we produced a filtered version of our message, devoid of most electrical noise.

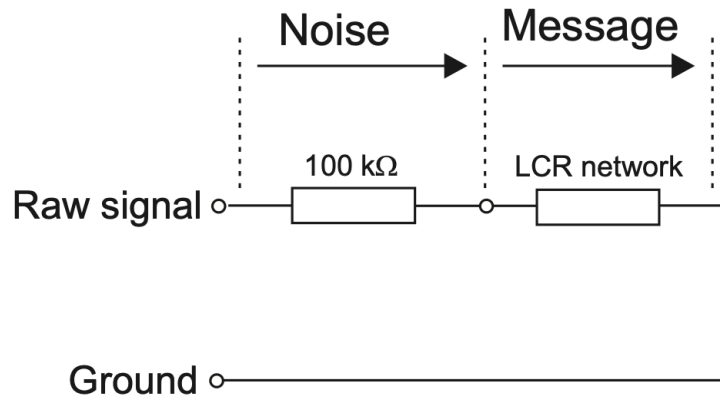


Figure 3: Signal filtering circuit [1]

By testing the suitability of manipulating impedances in circuits to filter out voltages of unwanted frequencies from a signal, we create solutions to a range of problems in the field of analogue electronics. Many engineering systems transfer electrical signals plagued by electrical noise caused by things like vibrations, thermal noise and electromagnetic interference. Signal filters such as that tested in this experiment can be crucial in removing electrical noise from modulated waves. LCR networks are also crucial in matching the impedance of a source and load in electrical systems, maximising power transfer between components and minimising signal reflections.

3 Method

3.1 Measurement of inductor resistance:

The inductor used in our LCR network was not perfect as it was made of a metal coil with its own resistance. So, we imagine this inductor as being in series with another resistor. A digital multimeter was placed across the inductor to measure this resistance.

3.1 Building the LCR network

To build our circuit, we used printed circuit boards that had on board resistors, capacitors and inductors. Using crocodile clips, we connected a capacitor in parallel to an inductor and resistor. The capacitance, C , was 66 nF, and we made R close to 500 Ω by connecting multiple resistors in series. We then joined the LCR network in series to a 1.0 k Ω resistor. By attaching a coaxial cable to a TG315 function generator, we connect it to both the circuit and a oscilloscope, which provides us with a visual representation of the wave made by the generator.

3.2 Measurement of voltages

With the TG315 disconnected, we measured the DC resistance, R_0 , of the LCR network, by placing a multimeter across it. Again, the zero error of the multimeter

and the absolute error from the limited digits were considered for this value. Then we connected 2 different multimeters across both the LCR network and the resistor attached in series with it to record V_1 and V_2 . For alternating sine wave voltages, the multimeters provide the root-mean square voltage across the relevant component.

3.3 Calculating inductance and other resonant properties

On the function generator, we set the root-mean square voltage of the applied voltage to 3V. By turning the frequency knob on the TG315 we vary the frequency of the applied voltage across the circuit. The value of Z_1 is calculated for different frequencies of applied voltages, ranging from 10 Hz to 10 kHz. By using excel to plot a curve of $\frac{|Z_1|}{R_0}$ against frequency, we determined the resonant frequency of applied voltage for which the impedance of the LCR network peaked. Using the value of resonant frequency, we determine the inductance of the inductor used in our network. The half power points and the bandwidth were determined from the curve which were then used to calculate the quality factor of the LCR network. By changing the value of the resistor in series with the inductor in the LCR network, we changed the amount of damping in the electrical signal across the network. We then obtained new plots of $\frac{|Z_1|}{R_0}$ against frequency for the different levels of damping.

3.4 Tuning circuit for signal filtering

A coaxial cable carried a modulated electrical signal at 5.0 kHz encoding a message in morse code. The signal contained electrical noise over a wide range of frequencies. Given that $\omega_0 = \frac{1}{\sqrt{LC}}$, where ω_0 is the resonant frequency of an LCR network, we can tune L and C so that the resonant frequency of the LCR network matches the frequency of the main signal. The inductor on our circuit board was fixed, so we chose a suitable combination of capacitors in parallel to tune the resonant frequency of the network as close to the desired value as possible. By connecting the network in series with a 100 k Ω resistor and connecting the coaxial cable carrying the signal across the circuit, we separate the message from the noise. By connecting the oscilloscope across the circuit and correctly setting its time-base and voltage scale settings, we obtain a visual representation of the filtered signal. By watching the oscilloscope trace in real time and comparing it to the morse alphabet given in the class manual, we decode the signal and obtain the hidden message.

4 Results

4.1 Resistance of inductor

The zero error on the multimeter was +0.2 Ω . Connecting the multimeter across the inductor and considering zero error, we measured its resistance to be $67.3 \pm 0.05 \Omega$.

4.2 Measuring impedance behaviour in response to varying frequency

For the first degree of damping, we set the resistance of the circuit as $R = 557.3 \, \Omega$ by combining the $390 \, \Omega$ and $100 \, \Omega$ resistors in series with the resistance of the inductor. Measuring the DC resistance across the LCR network using a multimeter yielded $R_0 = 558 \pm 0.5 \, \Omega$. The capacitance in the circuit is $66 \, \text{nF}$.

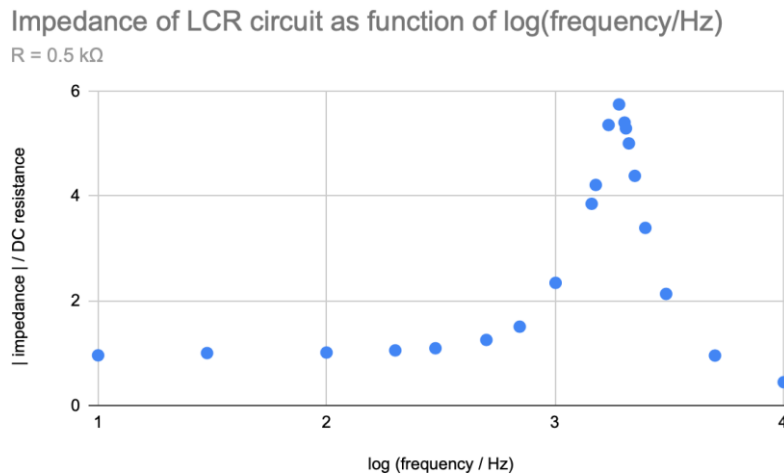


Figure 4: Impedance of LCR circuit as function of log (frequency), $R = 557.3 \, \Omega$

From the plot of $\frac{|Z_1|}{R_0}$ against $\log(f)$ shown in figure 4, we measure the resonant frequency to be $1900 \pm 5 \, \text{Hz}$. The two half power points are measured at $1410 \, \text{Hz}$ and $2240 \, \text{Hz}$ yielding a bandwidth of $830 \pm 5 \, \text{Hz}$. The Q factor is found to be 2.29 ± 0.02 by taking the ratio of the resonant frequency of the network with its bandwidth.

Finding inductance of network:

Using $\omega = \frac{1}{\sqrt{LC}}$, $\omega_0 = 1900 \, \text{Hz}$ and $C = 66 \, \text{nF}$, we find $L = 0.106 \, \text{H}$. Assuming the capacitor to have a tolerance of $\pm 5\%$, the uncertainty in L is $\pm 0.006 \, \text{H}$.

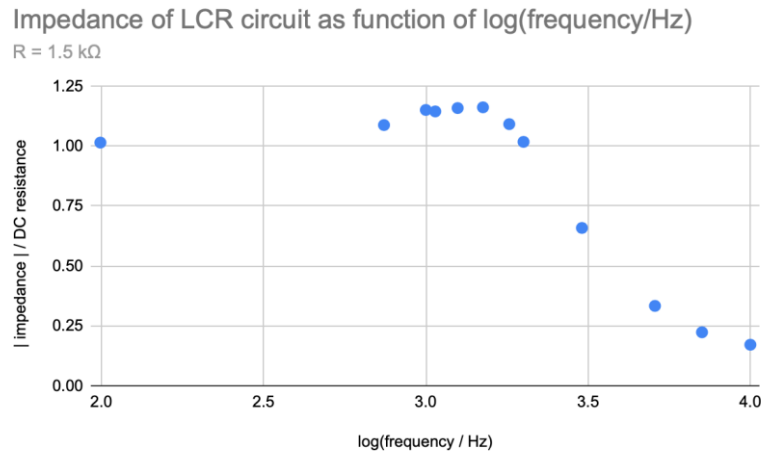


Figure 5: Impedance of LCR circuit as function of $\log(\text{frequency})$, $R = 1557.3 \Omega$

We increase the resistance of the LCR network by combining 100Ω , 390Ω and 1000Ω resistors in series with the 67.3Ω of the inductor to give $R = 1557.3 \Omega$. Placing a multimeter across the circuit gives a DC resistance of $1.55 \pm 0.005 \text{ k}\Omega$. The behaviour of the impedance in response to a changing frequency across the circuit is illustrated by the plot in Figure 5.

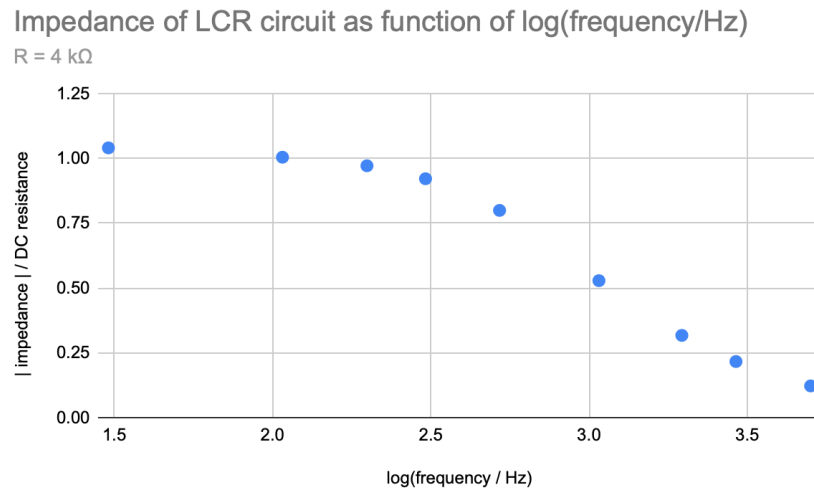


Figure 6: Impedance of LCR circuit as function of $\log(\text{frequency})$, $R = 4000 \Omega$

The damping effect in the circuit is amplified further by combining 100 Ω , 220 Ω , 390 Ω , 1000 Ω and 2000 Ω resistors in series with the resistance of the inductor to give $R = 3824 \Omega$. The impedance of the LCR network changes with respect to the frequency of the applied voltage as shown in figure 6.

4.3 Signal filtering

With an inductance of 0.106 H, the theoretical capacitance that makes a LCR network have a resonant frequency of 5.0 kHz is 9.56 nF. We joined a 3.3 nF and 6.8 nF capacitor in parallel to one another to form an effective capacitor of 10.1 nF. The only resistance in the circuit was that of the inductor at 67.3 Ω . By reading the oscilloscope signal across the LCR network, we deduced the message in morse code to be 'Witty message'.

5 Discussion

5.1 Measured values

As the level of damping in the LCR network increased from $R = 500 \Omega$ to 1500 Ω , the resonant peak of the $\frac{|Z_1|}{R_0}$ curve against $\log(f)$ decreased in amplitude and became broader. This is due to a more rapid loss of energy from the system in the highly damped network. When we increased the damping further to $R = 4000 \Omega$, there was no peak present in the curve and the impedance of the circuit reduced monotonically to 0 as ω increased.

The quality factor of the initial LCR network with $R = 500 \Omega$ was 2.29 ± 0.02 , calculated by dividing the resonant frequency of the curve by its bandwidth. Theoretically, Q is also equal to $\frac{\omega_0 L}{R}$ with ω_0 being the resonant frequency as 1900 Hz. This equation yields $Q = 2.27$ which is within the error range of our experimental value of Q , rendering it consistent with its theoretical prediction. The bandwidth of the LCR network was 830 ± 5 Hz: similar to bandwidths of networks typically used within audio applications.

The LCR network designed to filter the signal carrying morse code proved to be effective as it transformed the whole signal undecipherable on a oscilloscope to one that could be understood as a series of long and short pulses, with clear spaces between letter units. This confirms the suitability of LCR networks such as that tested in this experiment, to filter out electrical noise over a wide range of frequencies from main signals carried in a cable.

5.2 Errors

In our initial LCR network, we used a combination of resistors that should have given the network a resistance of $557.3\ \Omega$. However, using a multimeter to measure the DC resistance across the network gave $R = 558 \pm 0.5\ \Omega$. Similarly, in our second circuit, the theoretical resistance of the network $1557.3\ \Omega$ differed from the multimeter reading, $1550 \pm 5\ \Omega$. The inconsistencies of the measured resistances of the network compared to what they theoretically are may be caused by the resistance of the copper tracks and the capacitor used in the network. To counter this issue, one could use shorter and thicker metal tracks between components. Our circuit board also used metal springs in connecting components together: using low resistance metals for these would further reduce unwanted resistances in the circuit.

Another source of error was the fluctuation of the frequency of the voltage applied across the circuit by the TG315 function generator, giving frequency readings an uncertainty of $\pm 5\ \text{Hz}$. This uncertainty could be minimised by ensuring the power supply to the generator is clean and using shorter leads to connect the function generator to the circuit, reducing the chance of electromagnetic interference of the applied signal that could alter its frequency.

A source of systematic error in the measurement of resistances of the network and the inductor came from the zero error in the multimeter. The limited number of digits on the display also gave rise to an absolute uncertainty in these values.

6 Conclusion

A potential divider circuit was setup by connecting a LCR network (a capacitor in parallel to a series combination of an inductor and a resistor) in series with a resistor. Then by applying an alternating voltage across the circuit and measuring the separate voltages across the LCR network and the resistor in series with it, we recorded how the impedance of the network changed as the frequency of the applied voltage varied. When the resistance of the network was $557.3\ \Omega$, we showed the presence of a peak impedance at a resonant frequency of $1900\ \text{Hz}$: using this resonant frequency and the bandwidth of the network, we successfully found the quality factor of the network to be 2.29.

Then we showed that the resonant peak impedance of the resulting plot reduced in amplitude and became broader as the resistance within the network was increased, effectively acting as additional damping.

Finally, the use of an LCR network was tested in filtering a signal carrying morse code and proved successful. The message received and decoded was 'witty message'.

References

1. 'E7 Electrical resonance instructions – Lent weeks 5&6', 2025, Laboratory manual: Department of Physics, University of Cambridge.